

MAE 1001



Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2025/>

Prof. Kartik Bulusu, MAE Dept.

Graduate Teaching Assistant:

Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:

Nathan Janssen (Senior, MAE Dept.)

Olivia Falciani (Senior, MAE Dept.)

Shota Kakiuchi (Senior, MAE Dept.)

Tech support:

Rithik Gunuganti (CS Dept., MS student)



School of Engineering
& Applied Science

THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu

Seating Arrangement

Table 1 (8 students):

- Inti Stephenson
- Lucas Narvaez
- Naomi Kim
- Curtis Charlery
- Camryn Lorencovitz
- Stella Iyengar
- Shu Shu Wu
- Daris Dedic

Table 2 (8 students):

- Evan Chukwunonye
- Diego Flores
- Akina Pallera
- Sena Mute
- Cody Keel
- Matheo Valdivia
- Jessica Gil
- Kim Zamora

Table 3 (8 students):

- Cesar Tovar
- Dylan Nichols
- Kazi Afra Saiara
- Zachary Bogan
- Eli Ecret
- Addison Dooley
- Harper Barrios
- Andy Pham

Table 4 (8 students):

- Ryan Barks
- Nate Behabtu
- Irene Oh
- Hamza Chaudry
- Dennis Ventura
- Yahya Ahmed Ahmed Sadek Elsewedy
- Wassime Ait-Si
- Omar Mohammed A I Al Emadi

Table 5 (8 students):

- Teresa Cherian
- Tim Harrington
- Zaynah Asfour
- Novah Wali
- Grace Ephrem
- Nabek Ababiya
- Otto Erhart
- Maeve Carey

Table 6 (7 students):

- Paige Romanowski
- Daniel Zhang
- Brandon Alvarado
- Henry Steffen
- Lilia Hammadi
- Cadence Cassidy
- Aidan Willett

Table 7 (7 students):

- Saihaj Kaur
- Ali Ahmed Y A Al Malki
- Jocelyn Carrillo
- Alex Lagrant
- Malik Ali
- Chance Hwang
- Ethan Peralta

Table 8 (7 students):

- Owen Fritts
- Gael Merida Chavez
- Holt Perryman
- Mohammed Jabir H A Al Jabir
- Evan Agapiou
- Albert Wu
- Thomas Melotto

Table 9 (7 students):

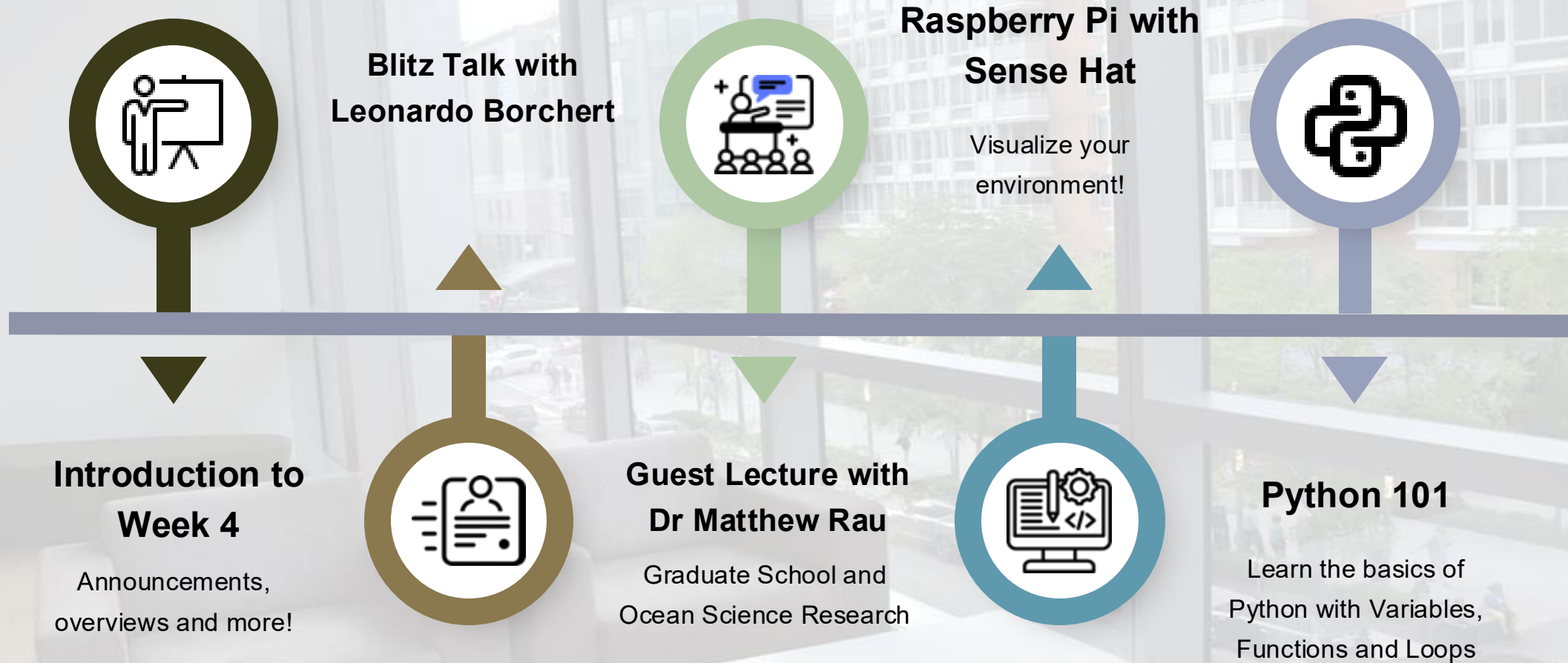
- Luz De La Cruz
- Alexander Lele
- Marthine Jean Pierre
- Ivan De La Cruz
- Tre Troy
- Johnny Lopez Bertrand
- Emre Ozkazanc



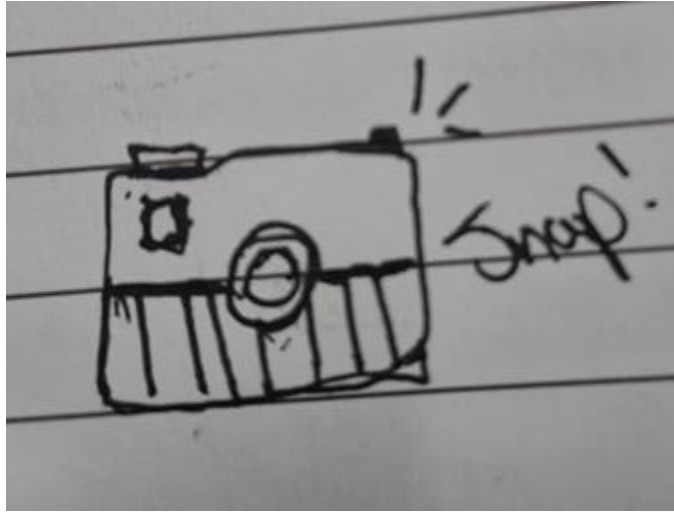
Road Map for Class 4

30-Minute Session Blocks Breakdown

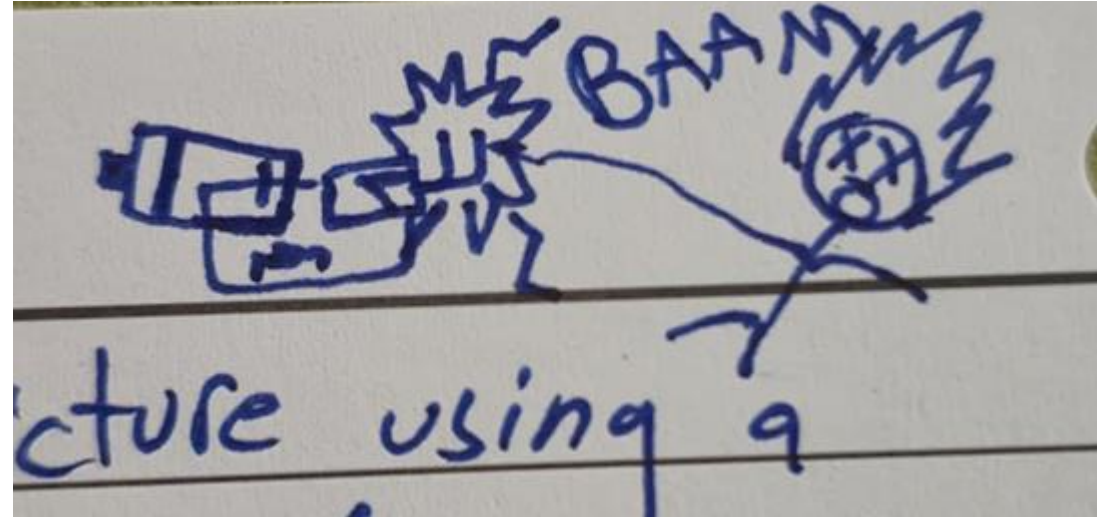
Photo Credits: Kartik Bulusu
Icon source: canva.com



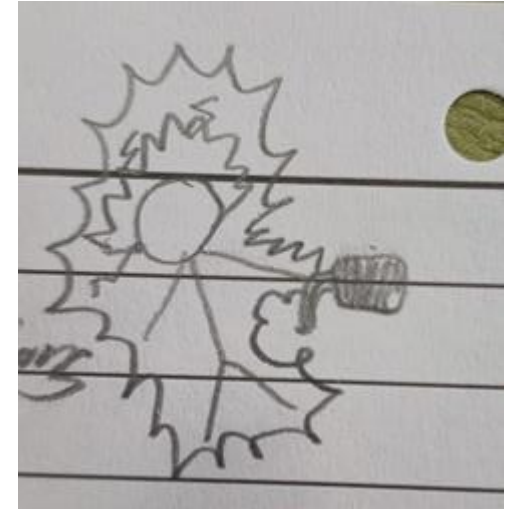
Week 3 - Best Doodles



Cadence Cassidy



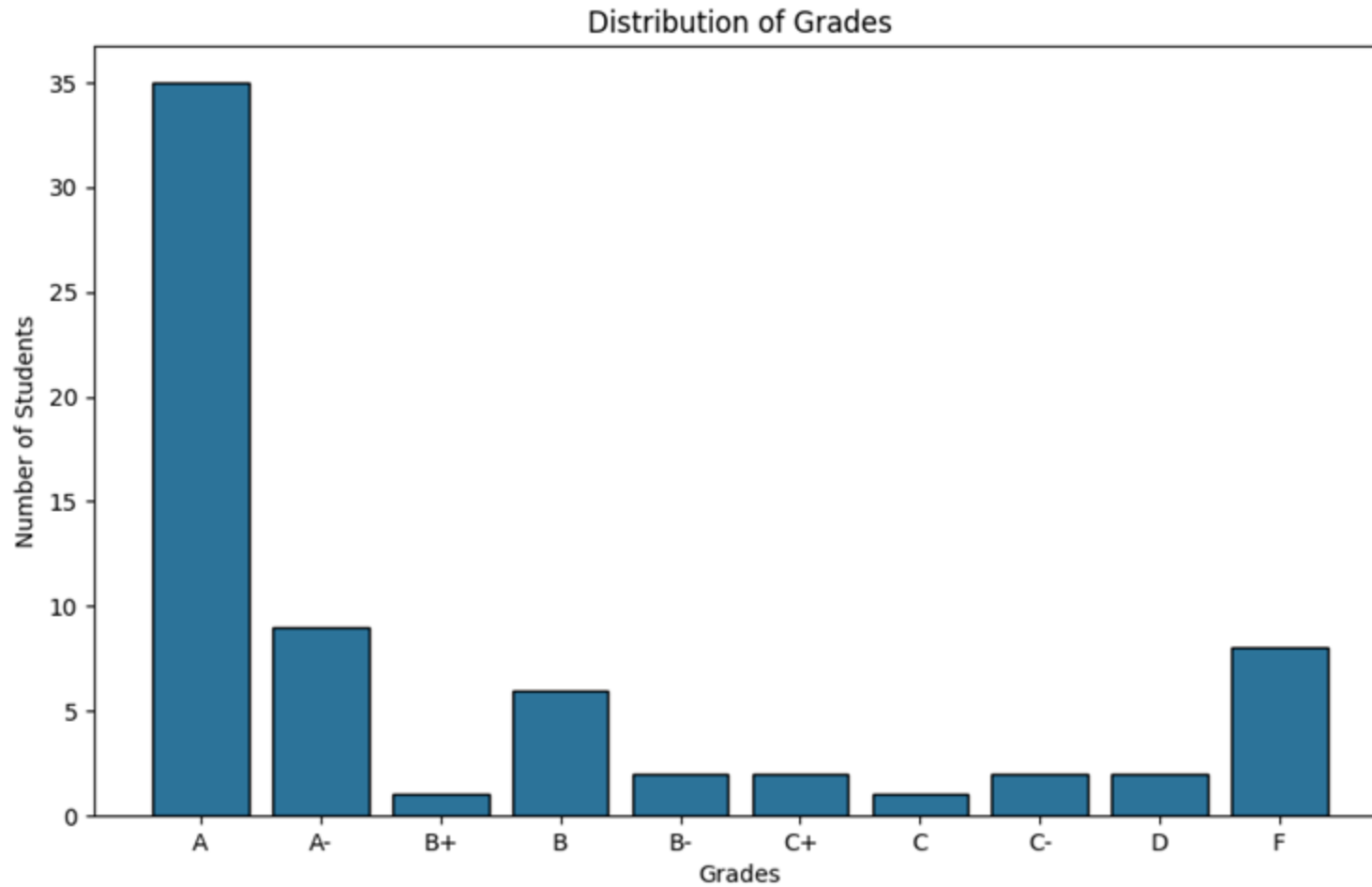
Zaynah Asfour



Malik Ali



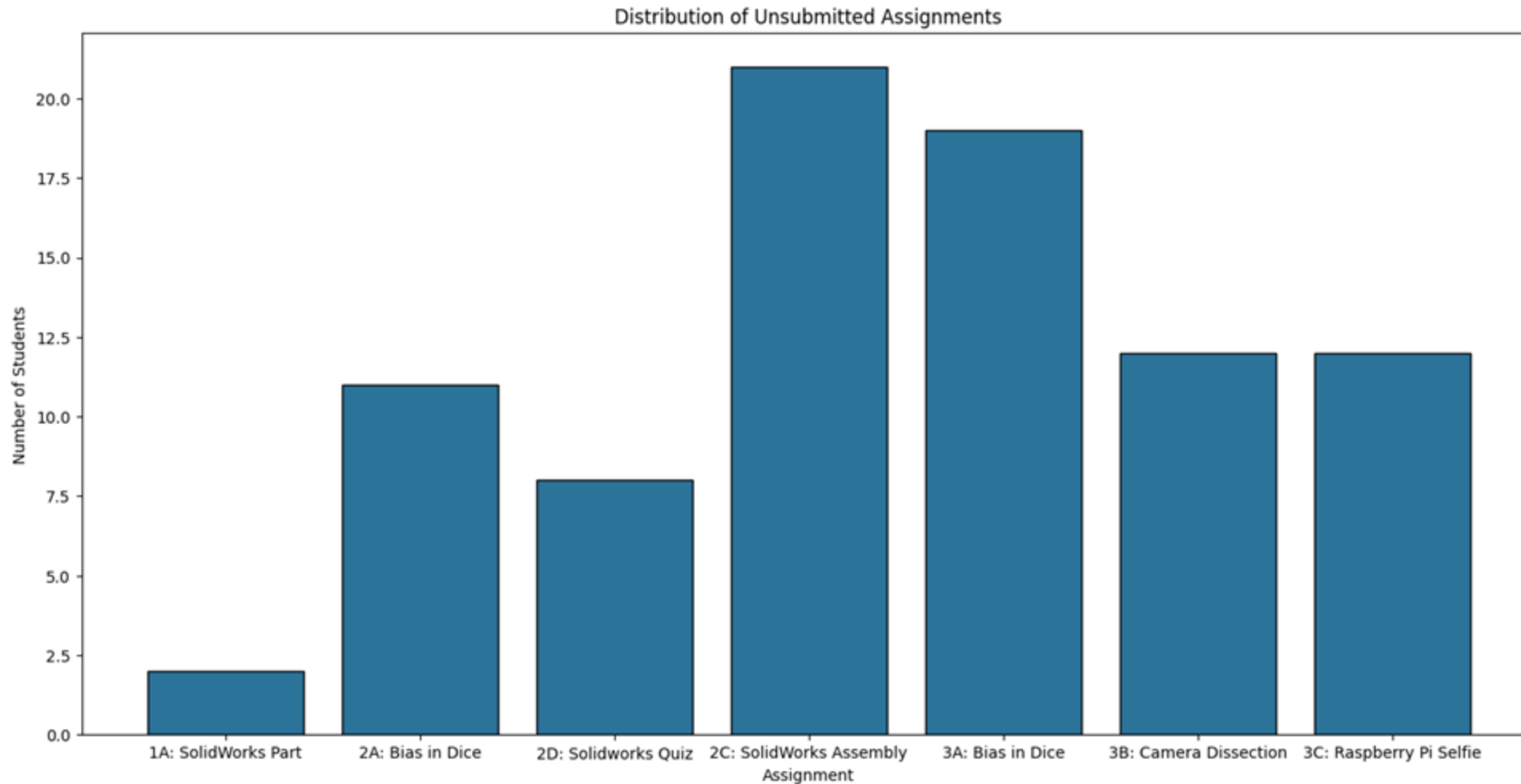
Week 4 - Grade Distribution



Score	Grade	Count of Students
90-100	A	35
84-89	A-	9
78-83	B+	1
72-77	B	6
66-71	B-	2
60-65	C+	2
56-59	C	1
52-55	C-	2
46-51	D	2
0-45	F	8



Week 4 - Assignment Distribution



GRADING SCALE

Solidworks Assignment: 30%

Quiz: 20%

Bias in Dice: 25%

Camera Labs: 25%



Blitz Talk with Leonardo Borchert



Senior in Mechanical and Aerospace Engineering

Email: leoborchert@gwu.edu

LinkedIn: [linkedin.com/in/leonardo-borchert-993212260/](https://www.linkedin.com/in/leonardo-borchert-993212260/)

Leonardo is a Mechanical and Aerospace Engineering student at the George Washington University. As Propulsion Division Lead for the GW Rocket Team and a three-time NIST CHIPS Act fellow, his work spans advanced propulsion and semiconductor research. He aspires to pioneer next-generation aerospace systems through industry and entrepreneurship.



Guest Lecture with Matthew J. Rau



Associate Professor of Mechanical and Aerospace Engineering

Email: matthew.rau@gwu.edu

Website: blogs.gwu.edu/matthew-rau/

Matthew Rau is an Associate Professor in the Mechanical and Aerospace Engineering Department at the George Washington University. He received his PhD from Purdue University in 2016 and was a National Research Council Postdoctoral Fellow with the Naval Research Laboratory from 2016-2017. Dr. Rau's research interests primarily consider multiphase flows in engineered systems (e.g. bubbly flows in processing equipment) to the environment (e.g. particle-laden flows). His group performs a variety of experimental techniques in the laboratory and develops and deploys novel measurement techniques in the field to study natural oceanic particulate matter.



Exit Ticket

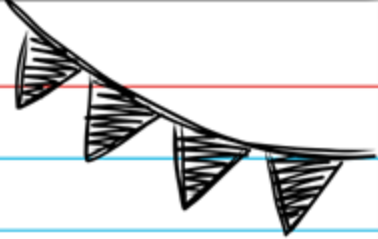
Name: _____

Three things you learnt today:

1. _____
2. _____
3. _____

Best Part of the Class: _____

Most Challenging Part of the Class: _____



Assignments for Week 4 - Due 11/20

**In-Class Lab 4A: Bias in Dice
Experiment - Trial 3**

In-Class Lab 4B: RPi Sense Hat Plot

**Assignment 4C: Week 3 Quiz - Guest
Lectures, Raspberry Pi and Python
on Blackboard**

